

Lab Report 5: Optical Qubits & Quantum Interference

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Experimental Setup Used: #1

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1 Abstract

This experiment explores the probabilistic nature of photons by demonstrating single-photon interference and its application in quantum information processing. Using spontaneous parametric down-conversion (SPDC) of a 405 nm pump photon, we herald the presence of an idler photon and direct the corresponding signal photon through a Mach-Zehnder Interferometer (MZI). By introducing a phase shift in one arm of the MZI, we observe oscillations in the coincidence counts, revealing the interference pattern of single photons. This behavior displays the principle of quantum superposition and enables the encoding of qubits in the photon's path degree of freedom. Our results demonstrate how linear optical elements can be used to implement fundamental quantum gate operations.

2 Introduction

The quantum nature of light is one of the most profound discoveries in modern physics. In the early 1900s, Albert Einstein postulated that light consists of discrete energy packets (photons) through his explanation of the photoelectric effect. This quantization, along with wave-particle duality as proposed by De Broglie in 1924, laid the foundation for quantum optics.

In this lab, we explore single-photon interference using spontaneous parametric down-conversion (SPDC) to generate correlated photon pairs. Detection of one photon (the idler) heralds the presence of the other (the signal), ensuring the system operates with single-photon

input. The signal photon is directed through a Mach-Zehnder Interferometer (MZI), where a controlled path-length difference induces a relative phase shift. By observing coincidence counts at the output, we reveal interference fringes, demonstrating quantum coherence and enabling insight into quantum logic operations.

3 Procedure

The experiment begins with a 405 nm pump laser set to 20–25 mW, which undergoes SPDC in a BiBO crystal to generate horizontally polarized signal and idler photons. The pump half-wave plate (HWP) is set to 0° , ensuring vertical polarization before the crystal. The idler photon is detected using a Single-Photon Avalanche Diode (SPAD) to herald the signal photon.

The signal photon is routed into the MZI. A piezo-controlled delay stage is used to introduce a variable phase shift in one arm. A linear polarizer in front of the MZI detector is set to 90° (horizontal transmission), matching the idler detector's polarization filter for optimal photon detection.

Before scanning the phase delay, background counts are measured with the laser blocked to account for accidental coincidences due to ambient noise. Once unblocked, signal, idler, and coincidence counts are recorded while incrementally adjusting the delay. The delay range and step size are chosen to capture multiple periods of the interference fringe. From

this data, we calculate the stage's step to distance calibration, determine fringe visibility, and observe how interference changes when one arm of the interferometer is blocked.

4 Results and Discussion

4.1 Interference Pattern from Idler and Coincidence Counts

In this part of the lab, we investigated the quantum interference in MZI using single photons. Our purpose was to demonstrate how a path-encoded qubit shows interference when both interferometer arms are open.

We started by measuring the idler and coincidence counts as we shifted the translation stage in steps of 2 from 0 to 40. By fitting a sine wave to our measured coincidence counts, we were able to obtain the period of the oscillations and extrapolate the step size in nanometers as 36.9nm per step. The following function, from the prelab, was used for fitting the data:

$$P = \frac{1}{2}(1 + \cos \delta) \quad (1)$$

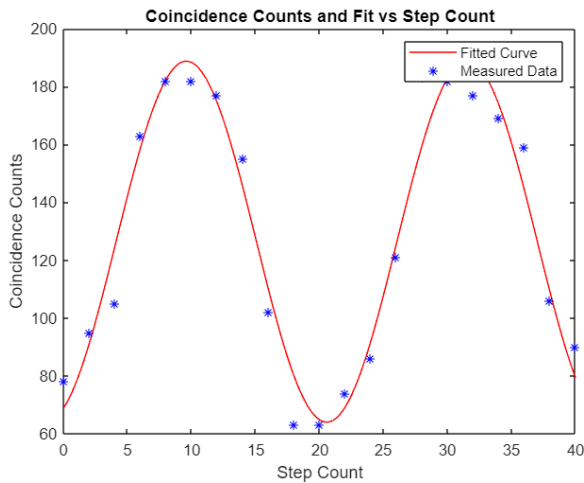


Figure 1. Coincidence Counts vs Step Count with a Sine Wave Fit

Figure 1 shows agreement between our data and the equations we derived in the prelab, with the fitted curve being the fitted equation. Besides using this in order to figure out the nanometers per step, this was also used in order to figure out the overall phase delay as shown in Figure 2.

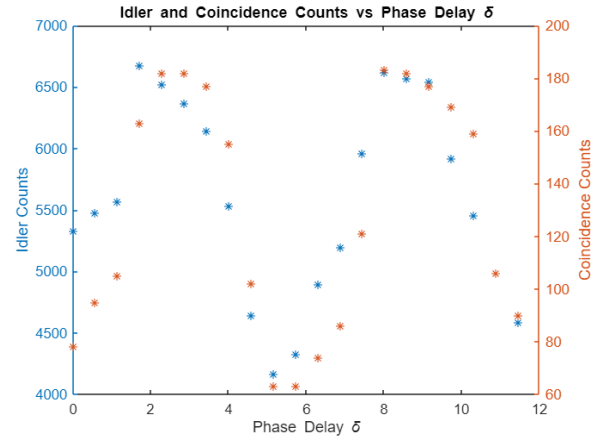


Figure 2. Idler and Coincidence Counts vs Phase Delay

We then measured the background counts and calculated accidental coincidence counts, and subtracted these from our data in order to get a more accurate dataset. Additionally, the phase delay was converted into nanometers to plot the corrected counts versus the path length difference.

$$N_{acc} = N_s N_t \Delta t \quad (2)$$

	Background Counts
Signal	2159
Idler	1270
Coincidence	0

Table 1. Background Count Measurements

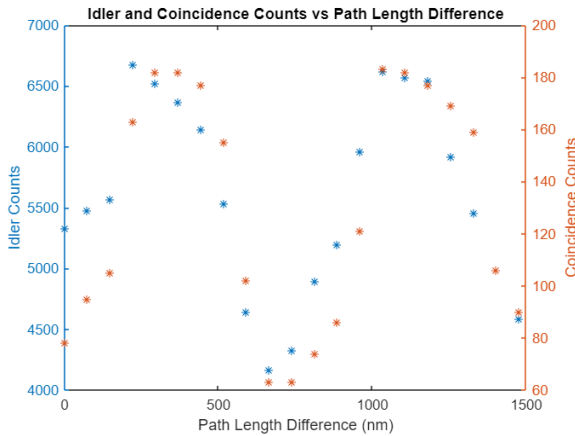


Figure 3. Idler and Coincidence Counts vs Path Length Difference

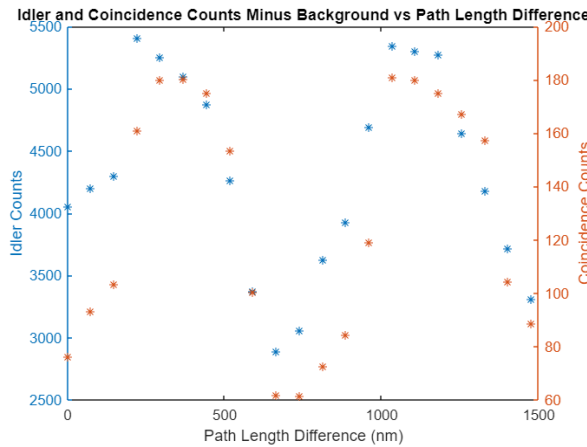


Figure 4. Idler and Coincidence Counts vs Path Length Difference with Background and Accidental Counts removed

The visibility for coincidence counts was measured to be roughly 0.5:

$$V = \frac{182-61}{182+61} = 0.498 \quad (3)$$

This makes sense because in the equation describing this interference pattern, as seen in equation 3, if the interference term (cosine term) is set to 0 as would be the case for destructive interference, then the equation would evaluate to 0.5. If it was set to 1 as would be the case for constructive interference, then the equation would evaluate to 1. Therefore, it makes sense that we see a visibility of 0.5 and it shows agreement between our results and theory.

We are able to see the interference in the signal despite only sending a single photon through the interferometer at a time because the photon, before being measured, is in a superposition of traveling down both paths, and the wavefunctions of both interfere with each other at the output of the interferometer. This creates a result that mimics classical interference despite there only being a single photon in the interferometer at a time.

4.2 Interference Pattern with one arm blocked

Blocking the upper arm (one without the translation stage) of the MZI, we repeated our measurement of the idler and coincidence counts as we varied the stage from steps 0 to 40. Figure 5 shows that we do not observe observations that would indicate interference, as we saw in the previous section. This is expected as, by blocking one arm of the MZI, we have prevented any interference between the two arms from occurring. Additionally, we see that both the idler and coincidence counts are roughly half of the average from the previous measurements. This is also expected as we are blocking half of the light in the MZI from reaching the output by blocking one of the arms.

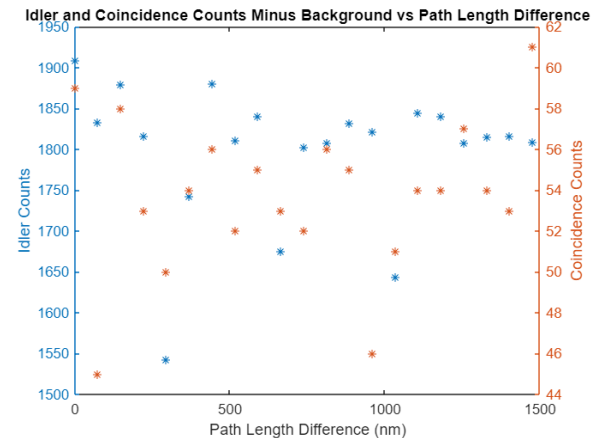


Figure 5. Idler and Coincidence Counts vs Path Length Difference with Background and Accidental Counts removed and the upper MZI arm blocked

4.3 Interference or No Interference?

Next, we adjusted the phase delay such that the SPAD counts were at a minimum within the interference pattern. Then, we recorded the counts, blocked the upper arm of the MZI, and recorded the counts again. This showed a change in the idler counts from 4133 to 3017 and the coincidence counts from 68 to 51. Generally, when we block half of the power, you would expect the counts to decrease by a factor of two, but they don't here.

This makes sense because while the upper arm was unblocked, the measured counts were subject to destructive interference, resulting in the counts being less than without interference. Then, when we block the upper arm, there is no more destructive interference, and the counts corresponding to the light from that arm increase. We just see a decrease in overall counts as half of the light in the MZI is blocked in the upper arm.

Counts	Unblocked	Blocked
Signal	12494	12700
Idler	4133	3017
Coincidence	68	51

Table 2. Counts with the upper path of the MZI unblocked and blocked

4.4 Qubits and Quantum Gates

The Mach Zehnder Interferometer in this lab acts can be connected to the quantum gates described during lecture. When the signal photon enters the MZI, it enters a state of quantum superposition where the lower path can be described by $|0\rangle$, and the upper path by $|1\rangle$; this can be described as a path encoded qubit. The phase delay introduced by piezo-controlled mirrors, acts as an arm phase operator.

$$\hat{Y}(\theta) = \begin{bmatrix} e^{i\phi_U} & 0 \\ 0 & e^{i\phi_L} \end{bmatrix}$$

Here ϕ_L , represents the delay δ , introduced by changing the value of the piezo-controlled motor, and ϕ_L is equal to 0, since there is no delay in the upper arm. Combining all the operations that occur, including the phase operator due to the LP, the Beam Splitter Operation when the photon enters the MZI, the Mirror Operator which changes the spatial dimension in of the arms, the arm phase operator, and the phase operator due to the MZI LP, we end up with the following:

$$|\psi'\rangle = \hat{Z}(\phi_2) \hat{B} \hat{Y}(\theta_1) \hat{M} \hat{B} \hat{Z}(\phi_1) |\psi_0\rangle$$

This equation can describe the output of the MZI.

4.5 Interference

Young's double slit experiment is a classic example of how a single photon can interfere with itself. In that experiment, a spatial interference pattern is demonstrated versus the interference pattern of an MZI, which requires adjusting the phase delay to see the oscillations

(pattern). However, they operate on the same principle. In both cases, a single photon is in a superposition of both paths simultaneously. This allows its wavefunction in both paths to interfere with one another, resulting in the interference pattern. This principle is the reason why both of these experiments still work with only a single photon.

5 Conclusion

In this lab we successfully demonstrated the principles of quantum interference and qubit manipulation using MZI. By encoding bits in the spatial path of photons, we observed interference fringes in both signal and coincidence counts as a function of phase delay introduced through a piezo-controlled stage.

Our corrected and normalized data shows oscillatory behavior in photon detection rates. The measured visibility of interference fringes provides evidence of superposition.

There are multiple things we can improve in our experiment. We could try to minimize background noise to increase signal to noise ratio. In addition, we could use a longer integration time to decrease statistical uncertainty. Furthermore, there might have been slight misalignments with equipment or tampering with optical paths that could introduce error in delay measurements.